Entity-conditioned nulls for anomaly detection in authentication logs

Abstract

Anomaly detection on enterprise security telemetry is usually posed as outlier detection against a pooled population: a model is fitted to all accounts and an event is flagged when it lies far from that fit. We argue and then measure that this asks the wrong question. An account that is permanently unusual is not thereby suspicious, and an account that has changed is suspicious whether or not it is unusual for the organisation. We therefore condition each null on the entity that produced the event, so that the reference set for an authentication is that account's own history rather than the population's.

Evaluated on the Los Alamos National Laboratory authentication corpus, where 549 labelled red-team events occur among 4.19 million scored events, a per-entity novelty test attains a precision of 15.7% (95% CI 9.0–26.0) at 10 alerts per analyst-day against a base rate of 1.31×10^{-4}, a lift of about 1,200. A structural census locates the mechanism: the categories the per-entity tests express are enriched 139- and 184-fold among labelled events, while the category a population-rarity test expresses contains none of them. Two categories that entity-scope tests also express — off-hours activity and volume bursts — are not enriched at all, which accounts for those detectors' null results without appeal to misspecification.

We report four negative results of independent interest. Combining the detectors' p-values, by Fisher's method with Brown's correction or by the Šidák-corrected minimum, does not exceed the best single component at any budget. Alerting whenever any detector ranks an event highly is strictly dominated at equal cost and superior at equal depth, at 3.7 times the alert volume. Dividing the budget in proportion to each detector's demonstrated quality, by a per-alert likelihood ratio fitted on a disjoint window, does not help either — and here the failure is bounded rather than merely observed: an exhaustive search over budget splits, choosing with the evaluation labels in hand, finds the optimum on the real campaign at the corner, the whole budget to the best single detector. The quantity that decides whether dividing a budget can pay is the overlap between two detectors' detections, which is 74.6% between the two strongest here and 0% in the one configuration where a split does win. And a population-scope model retains a capability the entity-scope framework lacks, so the two scopes are complementary rather than ordered.

Keywords: anomaly detection; multiple testing; conditional inference; false discovery rate; intrusion detection; decision theory.

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1. Introduction

1.1 The operational problem

An enterprise authentication log records who authenticated to what, from where, and how. A single record from the corpus studied here carries nine fields: a timestamp, a source and destination account, a source and destination computer, an authentication type such as Kerberos or NTLM, a logon type such as Interactive or Network, an orientation, and whether the attempt succeeded. A field with no interpretable value is written as a literal `?`, which is a distinct state from absence and is treated as one throughout.

Two features of this setting determine what any statistical method can achieve on it.

The first is volume against rarity. A mid-sized organisation generates tens of millions of authentications a day as ordinary business: service tickets, screen unlocks, share mounts, scheduled jobs. An intrusion touches tens of events. On the corpus used here, 42.2 million events were scored over seven days and 549 of them are labelled, a base rate of 1.30×10^{-5}. Writing π for that rate and α for a per-event false-alarm rate, the share of alerts that are real is

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P(intrusion | alarm) = π / (π + (1 − π)·α),
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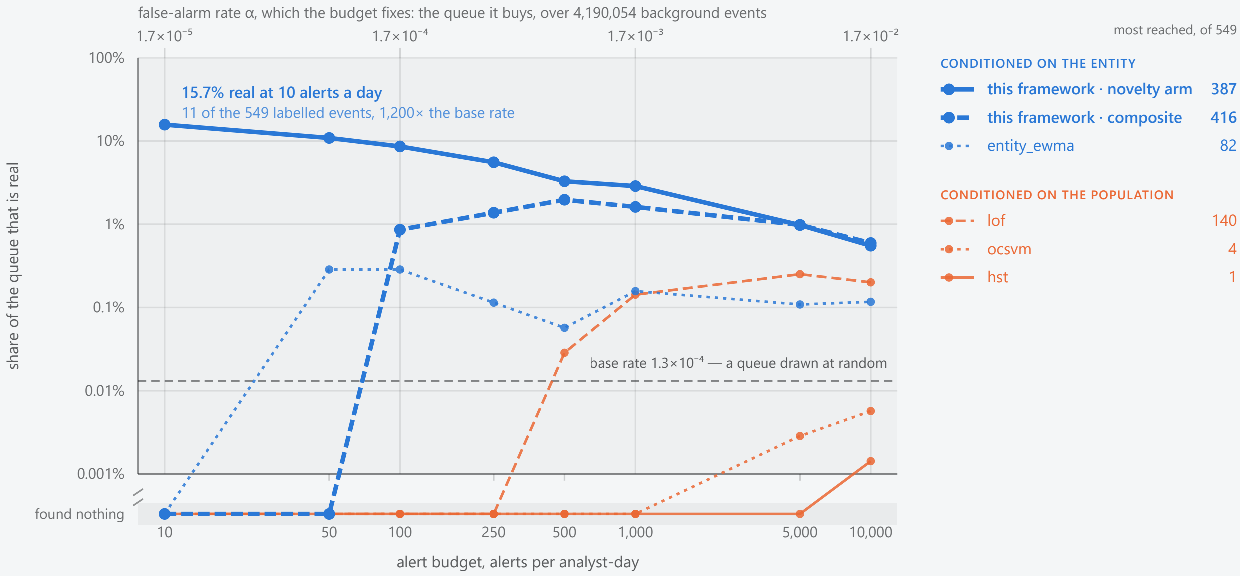
so half a queue is real only at $\alpha \approx 1.3 \times 10^{-5}$, or about 78 false alerts a day. At $\alpha = 10^{-3}$, an operating point typical of published detectors, the same corpus yields approximately 6,000 false alerts a day against 78 real ones. This is Axelsson's base-rate argument [1] applied to our own data, and its usual consequence is that the binding constraint is the suppression of false alarms rather than the recognition of intrusions.

Share of the alert queue that is real, at each alert budget

4,190,603 events scored over 7 days · 549 labelled · a base rate of one in 7,633. Grouped by the scope of the null each method tests.



The comparison is a function of the budget, not a point. Every line is measured on the same slice: 4,190,603 scored events over 7 days, of which 549 are labelled, a base rate of one in 7,633. Each point is one method at one measured budget. A method that found nothing at a budget is drawn on the *found nothing* row below the axis break rather than omitted, so a flat line there reads as measured rather than as missing; the methods leave that row at six different budgets, which is the comparison this figure exists to make.

Colour is the scope of the null, not who wrote the method. Blue judges an event against the history of the account that produced it, orange against the pooled population (§1.2). `entity_ewma` is a comparator drawn in blue because it keeps a per-entity mean; this framework's arms are the heavier strokes in that family, and every method's scope is recorded in the curve file. A method's own α is lower than the budget's by the detections it found, at most 16% at 10 alerts a day and 0.6% at 10,000.

The figure draws the four comparators with the most detections over the budget range, so the choice does not flatter this framework; the other four reached too little to plot, and the table below lists all eight that were run. *Every reading here is an upper bound on this framework's disadvantage and a lower bound on its precision.* The corpus has no true negative class, so an unlabelled alert on genuine but unrecorded malicious activity counts as a false alarm (§12.5). The baselines read a fixed-length numeric vector per event and only `entity_ewma` holds per-entity state; they are not held to R2 or R4 (§12.4).

The figure measures that trade-off across the operating range rather than asserting it at one point. Each line is one detector as the budget rises from 10 to 10,000 alerts a day, plotting the share of its queue that is real against the budget that bought that queue. The false-alarm rate is not given an axis of its own: a budget of b buys the same $b \times 7$ alerts whichever method spends it, so α follows from the budget and is read off the scale above the plot. The comparators are the strongest four of eight reference implementations run on the same events, and the lines are coloured by the scope of the null each one tests rather than by who wrote it.

The table beneath it fixes the budget and reports what each method actually did, at two operating points and for every comparator that was run, including the four the figure has no room to draw. It is placed this early on purpose: the figure answers how the comparison moves with the budget, and a reader arriving at the paper wants first to know what the numbers are.

METHOD	JUDGED AGAINST	100 ALERTS / ANALYST-DAY			1,000 ALERTS / ANALYST-DAY		
		FOUND	PRECISION	RECALL	FOUND	PRECISION	RECALL
this framework, novelty arm	its own history	60	8.6%	11%	201	2.9%	37%
this framework, composite	its own history	6	0.86%	1.1%	113	1.6%	21%
entity_ewma	its own history	2	0.29%	0.36%	11	0.16%	2.0%
lof	the population	0	0	0	10	0.14%	1.8%
ocsvm	the population	0	0	0	0	0	0
hst	the population	0	0	0	0	0	0
rrcf	the population	0	0	0	0	0	0
eif	the population	0	0	0	0	0	0
iforest	the population	0	0	0	0	0	0
pca	the population	0	0	0	0	0	0

Every method that was run, at two budgets. A budget of b alerts per analyst-day buys a queue of $b \times 7$ alerts over the 7 days scored, whichever method spends it. *Found* is labelled events reached, *precision* the share of the queue that is real, and *recall* the share of the 549 labelled events reached.

Most of the comparators reach nothing at all: seven of the eight at 100 a day and six of the eight at 1,000 a day. A zero is measured, not missing — every method read the same events, which is why the figure above needs a row below its axis.

This framework's novelty arm reaches 11% of the campaign at 100 alerts a day and 37% at 1,000, against a base rate of 1.3×10^{-4} — a lift of 650× and 220×.

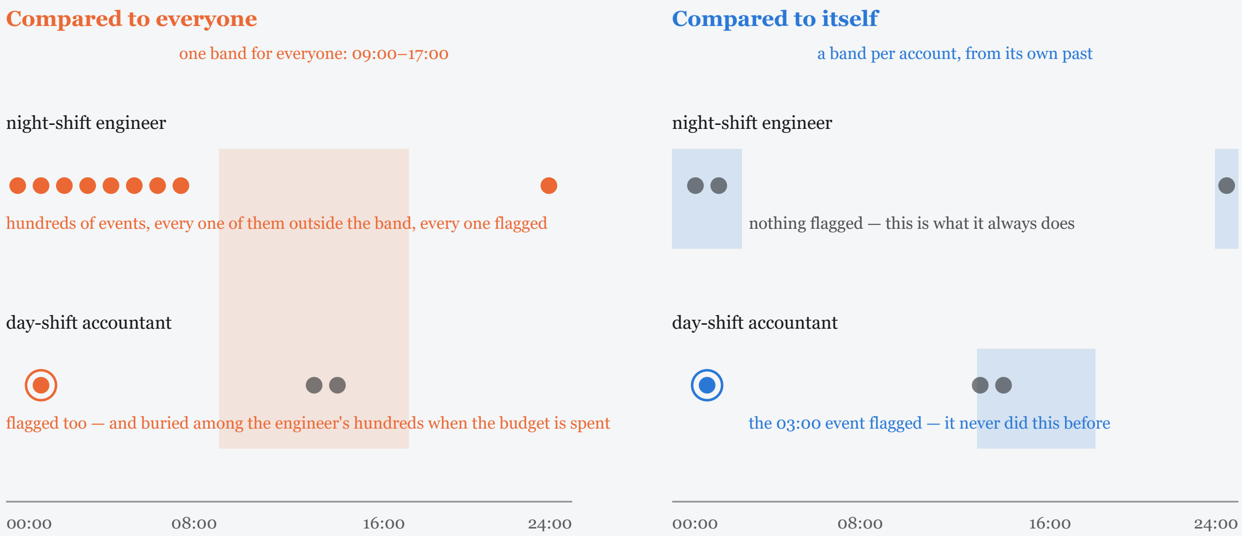
Recall counts what the red team wrote down, not what happened. The label file records an authorised exercise rather than annotating the log exhaustively, so unrecorded malicious activity counts against precision and is absent from the recall denominator (§12.5): both columns are lower bounds. Only `entity_ewma` holds per-entity state, and no comparator is held to R2 or R4 (§12.4).

Two readings, and the second is not in this framework's favour. Its per-entity novelty arm reaches labelled events at every budget, 11 at 10 alerts a day where every comparator reaches none; its own combined verdict does not: at 50 alerts a day it reaches none where a per-entity moving average reaches one, which is section 5.4's finding arrived at from the other direction. The subset measured here carries a base rate of one in 7,633, ten times the full corpus's one in 76,901, so every precision on the figure is easier than the arithmetic above demands. The conclusion is unchanged: suppressing false alarms to the rate the base rate demands is not this framework's binding constraint, because it already achieves it. Recall is.

The second is that the consumer of the output is a person with a fixed working day. A security operations centre triages a bounded number of alerts per shift because each costs a human minutes to hours. That bound is a staffing constant, not a tuning parameter, which is why every comparison in this paper is made at a matched number of alerts per analyst-day rather than at a matched threshold: two methods thresholded alike may emit volumes differing by three orders of magnitude, and the comparison would then be between their volumes.

The labels deserve a third remark, because they shape every figure. A red team is an authorised offensive exercise conducted by people who record what they did. The label file is that record: it is not an exhaustive annotation of the log. Two consequences follow and are carried throughout. Genuine malicious activity that the red team did not record, or that was not theirs, is counted as a false positive, so every precision figure in this paper is a lower bound. And the recall denominator is what was written down rather than what happened.

1.2 Why a population-scope null asks the wrong question



The reference moves with the account. Both panels hold the same two accounts and the same events. On the left one band serves everyone, and the failure is not that the accountant's 03:00 event goes unflagged — it is flagged. It is that the night-shift engineer is flagged on *every event it ever produces*, so the day's alert budget is spent on accounts that are behaving exactly as they always have, and the one event that matters is buried among them. On the right each account is judged against its own past: the engineer's night work is unremarkable, and the accountant's 03:00 event is the only thing left to look at. Nothing about the events changed, only what they were compared against.

Consider a systems administrator who works nights and authenticates to database servers no one else touches. A model fitted to the population places her far from its centre on every event she generates. She is a persistent outlier and she is not an intruder. Conversely, an account whose behaviour has just changed completely may still sit inside the population's bulk, because the values it has moved onto are ordinary for other people.

Both errors have the same cause: the question "is this event unusual for this organisation" is not the question an analyst needs answered, which is "has this account departed from its own behaviour". Formally, the two differ in the reference set against which an observation is judged exchangeable — the population of all accounts, or the history of one. Conditioning on the entity is the whole of the commitment this paper makes, and section 5.1 shows it is where the discriminating information on this corpus lies.

The distinction is not merely conceptual. It also determines what a fixed budget buys: a population-scope model spends its budget on whichever accounts are furthest from the mean, and those accounts are the same ones every day.

1.3 Contributions, and what is not claimed

We contribute: a family of entity-conditioned nulls over heterogeneous log fields that requires no advance declaration of a field's type, cardinality or value set (section 3.2); a measurement of what per-entity conditioning is worth at matched analyst-day budgets, with intervals (section 5.2); a structural census that identifies which properties of an event carry the discriminating information, and which do not (section 5.1); and per-mechanism sensitivity against a corpus of planted attacks designed to hold population rarity out (section 5.3); and a bound on what any allocation of a fixed alert budget across detectors can achieve, together with the statistic that decides whether dividing one pays at all (section 5.5).

We do not claim state-of-the-art detection. Precision reaches 15.7% at the tightest budget studied, which is good against a base rate of 1.31×10^{-4} and not good enough to act on unattended. We do not claim that combining evidence across detectors helps: four combination and allocation rules are measured, none exceeds its own best component at equal cost, and on the real campaign no division of a fixed budget does either, including one chosen with the labels in hand (sections 5.4 and 5.5). We do not claim that population-scope conditioning is useless; it retains one capability this framework lacks (section 5.7). And we do not claim generalisation beyond authentication telemetry of this shape, nor to the full population: the figures below are measured on entity subsets, for reasons section 2.2 states.

The framework is stated as six requirements, referred to throughout by number.

	REQUIREMENT
R1	A verdict is a function of the event and of persisted state only, so scoring is reproducible from the record and independent of batching.
R2	No component requires advance knowledge of a field's type, cardinality or value set. One field must be declared to be the entity, and one the timestamp; nothing else.
R3	Abstention is an outcome. A component with no basis for an opinion says so, and does not contribute a neutral value.
R4	Scoring is deterministic: no wall clock, no randomness, and a canonical total order before every floating-point reduction.
R5	A verdict carries the evidence needed to reconstruct it.
R6	Alert volume follows from a stated exchange rate between a missed intrusion and a wasted investigation, not from a quota.

2. Data and evaluation design

2.1 Authentication telemetry and its labels

We use the Los Alamos National Laboratory comprehensive cyber-security events data set [2], released CC0, and specifically its authentication log: 1.05 billion events over 58 days from 12,425 users and 17,684 computers, with a labelled red-team exercise embedded in it.

The unit of analysis is a human account. Machine accounts, which the corpus writes with a trailing `$`, together with `SYSTEM` and `ANONYMOUS LOGON`, are excluded: they are not individuals, and an ethogram of an individual is what this framework builds. The restriction costs no labels, because all 104 accounts the red team touched are human accounts.

The label file carries 749 rows, of which 715 are distinct — 34 are exact duplicates — naming 104 accounts. Its structure is highly concentrated: 93.6% of rows share a single source computer, and two of the seven scored days carry 482 of the 700 post-boundary labels. The effective sample size is therefore far below 549, a point section 6.2 returns to.

Two properties of the corpus bear on specific detectors. Distinct human-user counts follow a weekly cycle, roughly 12,800 on weekdays against 5,400 at weekends, with daily volume between 13.1 and 25.0 million events; a null over an account's hour-of-day or event count must absorb that. And failed authentications appear in the corpus only for accounts that succeeded somewhere in the data set, so the failure population is conditioned on eventual success. That is a property of the archive and not of a live stream, and the success field is one the detectors score, so it is a genuine selection artefact; we report it here and revisit it in section 6.2.

2.2 Windows, entity population and sampling

The first seven days ($t < 604{,}800$ s) are a burn-in window: events are scored, so that state warms under exactly the code path that scoring uses, but nothing is emitted, counted or alerted on. Two quantities are fitted on this window and frozen at its boundary — a between-detector covariance and a conformal calibration (section 3.3) — and nothing fitted there is ever updated by an event it is later used to score.

The window was chosen before any end-to-end measurement was taken and is recorded in the run metadata by the commit that fixed it. It costs the 49 labelled events that fall inside it, on days 1, 2, 5 and 6. We note the provenance because it is the paper's defence against the charge that the split was chosen after seeing results: it was not, and the record shows it.

The scoring window is $t \in [604{,}800, 1{,}209{,}600)$ s: seven days, days 7 to 13 inclusive.

Four sampling designs appear below. Three are entity samples, taken because a fixed alerts-per-day budget is a far harder target on 42.2 million events than on 4.2 million, and because a population-scope quantity computed on a sample is a statement about the retained accounts. Every labelled account is retained in every sample, which inflates the labelled share and is why each design's base rate is stated rather than assumed.

DESIGN	SELECTOR	SCORED EVENTS	LABELLED	BASE RATE
full	none	42,218,530	549	1.30×10^{-5}
`r11`	1 entity in 16, residue 11	4,190,603	549	1.31×10^{-4}
`inj`	1 entity in 16, residue 7, plus planted attacks	4,494,396	1,405	3.13×10^{-4}
`holdout-r7`	days 7–8 of the residue-7 corpus	1,427,225	262	1.84×10^{-4}

Because labelled accounts are exempt from sampling, the residue-7 and residue-11 corpora share their labelled events and differ only in background and window. A result reproduced across them is therefore a check on sensitivity to the background population, and not an independent replication; we describe it as the former.

2.3 Planted attack mechanisms

The real campaign supplies one uneven mixture of mechanisms, which cannot separate "this detector cannot express this mechanism" from "this corpus barely contains it". We therefore plant synthetic attacks of six named kinds into a held-out copy of the corpus, eight victim accounts per kind, entirely inside the scoring window. Victims are disjoint from every account the real labels name, so the account alone determines which ground truth an event belongs to.

Each kind isolates one premise. **Credential spray** sends an account to many destinations it has never used inside ten minutes. **Lateral movement** walks a connected path of hops, each sourced from the previous hop's destination, so that every computer is real and only the pairings are new — a mechanism no test scoring fields independently can express. **Off-hours access** uses only values the account already uses, twelve hours from its own established window. **Privilege escalation** introduces an authentication type absent from the account's history at its usual hour and host. **Low-and-slow** exfiltration uses familiar values only, with a modest sustained volume increase. **Account takeover** changes destination, authentication type, logon type and hour at once, as an upper bound.

The design decision that matters statistically is the choice of planted values. A planted value is the *most population-common value the victim has never used.* Sampling the vocabulary uniformly instead plants values that are rare population-wide, and then a population-rarity test detects the planted attacks for the wrong reason; that error was made in an earlier version of this corpus and is visible in its measurements. Choosing the most ordinary unused value tests per-entity novelty with population rarity held out, which is the only construction under which the per-mechanism table separates the two questions this paper is about.

Section 5.3 reports where that construction succeeds and where it cannot, which depends on the cardinality of the field being substituted.

Two limits on how a per-mechanism rate may be read. Value choice is deterministic given the corpus, so every victim of a kind receives the same planted values: the events of one kind are not independent draws and the effective sample size is nearer the eight victims than the event count. And sensitivity against planted labels is a claim about whether a test responds to a mechanism, not about detecting an intrusion. The two ground truths are reported side by side throughout and are never summed.

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3. Method

3.1 The entity-conditioned null

Write an event as a set of field–value pairs together with an entity *e* and a time *t*. For each field the framework maintains a per-entity summary of that entity's history, decayed exponentially in event time with a half-life of seven days, and asks a question of the form

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H₀ : this observation is exchangeable with e's own decayed history for this field.
```

Six such questions are asked, each returning a p-value under its own null, or abstaining. Nothing about the construction is specific to authentication; what makes a question available is the inferred kind of the field, not its name.

Novelty asks whether *e* has taken this categorical value before. The predictive is Dirichlet–multinomial with concentration a over the decayed counts, so a value seen n_v times out of n receives roughly (n_v + a)/(n + a(K+1)) and a first-ever value receives the reserved mass a/(n + a(K+1)). With a = 1 and n ≫ K that reserved mass is approximately 1/n, which is the detector's most important property and its principal weakness: **the p-value for a first-ever value is set by the length of the account's history rather than by how surprising the value is**, so a quiet account cannot produce an extreme p-value however anomalous its behaviour. Section 6 returns to this.

Novelty rate exists because of that weakness and asks a scale-free question instead: is *e* producing first-ever values faster than it historically does? Over an hourly window of m events, the count K of first-ever values is referred to a Beta-binomial predictive, K ~ BetaBinomial(m, a, b), where the Beta carries the posterior over *e*'s own rate of producing novelty. Carrying that posterior rather than a point estimate is what stops a second novel value from an account with twenty events of history being called overwhelming.

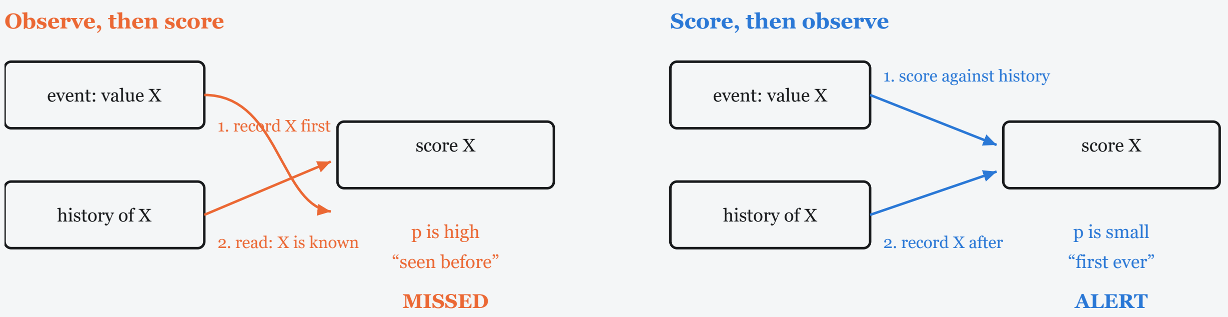
Timing asks whether the hour is unusual for *e*, estimating *e*'s time-of-day density on the 24-hour circle by von Mises kernel density estimation, maintained as a truncated Fourier series whose harmonic weights r_h = I_h(κ)/I₀(κ) are the kernel's own Fourier coefficients. The concentration κ follows from a bandwidth stated in hours, here 1.5.

Volume asks whether *e*'s activity in the current period is unusual for *e*, referring the period's event count to a Gamma–Poisson predictive fitted on *e*'s completed periods. The predictive is deliberately over-dispersed relative to the Poisson, which is what makes it tolerant of ordinary bursts and, as section 5.3 shows, blind to sustained small increases.

Pairing asks whether *e* has combined two values before. A pair is addressed as a value of a synthetic composite field, so novelty's estimator scores it unchanged; the question is per-entity, not population.

Marginal is the one population-scope question retained: is this value rare across the whole population, under decayed population counts with the same Dirichlet form? It is present because a comparison needs it, and because — as section 5.3 shows — it turns out to detect something the entity-scope questions do not.

Each detector may score several fields of one event; the arm for that detector takes the most extreme, and where the detector combines fields internally it applies a Šidák correction for the number tested.

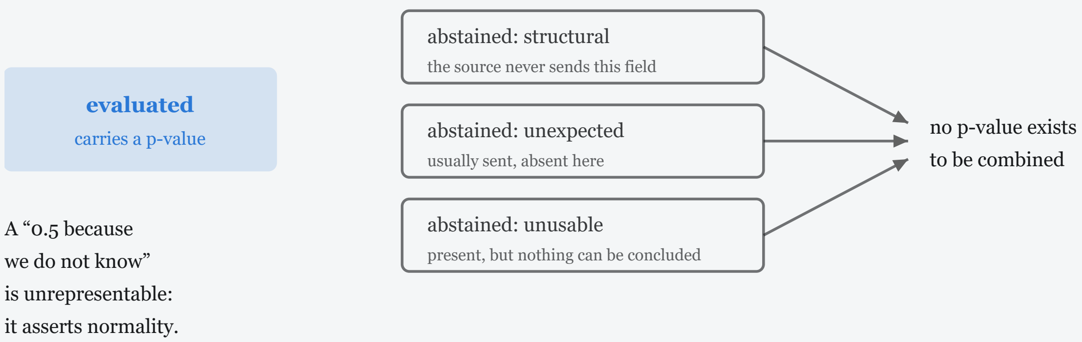


Both arrangements return a number. Only one of them is measuring anything.

The failure is silent, which is what makes it dangerous. Update state before scoring and a first-ever value is already a known value by the time it is judged, so novelty detection dies while still emitting plausible p-values. Nothing in the output announces it. The framework therefore makes the ordering unexpressible rather than conventional: `Score` holds no means of writing, and the update it computes does not exist until it has produced it.

Ordering is fixed and is the framework's leakage control: every detector scores against pre-event state, the combination is computed, and only then are observations committed and the field registry updated. A first-ever value is therefore still novel at the moment it is scored. The same rule applies at the window level, and it cuts the other way too: because an event updates e's state after being scored, a sustained intrusion progressively enlarges its own reference set. Section 6 treats this as a live threat rather than a solved problem.

3.2 Inferring what a field is, and declining to answer



Declining is not the same as reporting nothing unusual. A verdict has four states and only one of them carries a p-value, so a detector with no basis for an opinion cannot express a confident middle value — the field is unreachable. An abstention reduces the degrees of freedom of the combination rather than contributing to it. This is what lets the system degrade honestly instead of inventing findings when a source changes shape.

R2 requires that no component be told a field's type, cardinality or value set. A registry observes values as they arrive and settles each field into one of five kinds — categorical, boolean, discrete, continuous, identifier — after which the detectors iterate the registry rather than a list of field names. Onboarding a new log source is therefore configuration rather than code, and the one thing that must be declared is which field carries the entity.

The identifier kind carries the weight here. A field whose values are almost all distinct — a session GUID, a request id — is unbounded by construction: every value is a first-ever value, so a novelty test on it fires on every event and, being the most extreme thing in the queue, consumes the entire budget. The registry classifies such a field as an identifier and detectors decline it. That is not a heuristic convenience; without it the framework's headline arm is a random-number generator with a large p-value.

R3 makes abstention an outcome rather than a neutral score. A detector with no basis for an opinion — the field is absent, or present but not interpretable, or the entity has no history for it — says so, and the abstention reduces the number of tests combined rather than contributing a p-value of one. The distinction matters at this base rate: a neutral value entered into a combination over six tests is not neutral, it is evidence of ordinariness, and in Fisher's sum five such values will bury one informative test.

Values are handled as text and each inferred kind supplies its own scoring representation, so no field is cast to a numeric type on the strength of the first values seen.

3.3 Combination, calibration and allocation

Given J evaluated p-values for one event, four rules for turning them into one decision are measured. All four are defined here so that section 5.4 can report outcomes without defining machinery.

The composite is Fisher's method [3], $X^2 = -2\,\Sigma \ln P_j$, referred to $\chi^2(2J)$, with Brown's correction [4] for dependence between the detectors: the burn-in covariance of the $-2 \ln P_j$ supplies a scale c and effective degrees of freedom f, and X^2/c is referred to $\chi^2(f)$.

Brown's correction requires a positive variance estimate for X^2, and on this corpus it does not always get one. With six detectors the burn-in covariance implies $\mathrm{Var}[X^2] = -27.5$, which no joint distribution of the statistics can produce. When that happens the correction is not applied and the combination degrades to plain Fisher, and the run records that it did. The degradation is the specified behaviour rather than a repair invented for this paper, and it matters that it is stated: every composite figure below is Fisher's statistic without Brown's correction. The negative estimate is itself informative, and section 6 reads it.

The corrected minimum takes the smallest p-value and corrects it for multiplicity by Šidák [5]: $P = 1 - (1 - \min_j P_j)^J$. It asks whether *any* detector found the entity out of character, which is closer to the question the framework poses than whether the evidence is jointly unusual.

The union asks the same question with the p-value scale removed. Each detector ranks the day on its own p-value, an event takes the best rank it attains in any detector, the union is deduplicated on event identity, and ties break on that identity rather than on a p-value — rank collisions are structural, since every detector has a rank 1, and breaking them on log P would hand each collision to whichever detector's numbers happen to be smallest. It is charged two ways: **at equal cost**, truncated to the same alerts a day every other arm gets, and **at equal depth**, every detector keeping its own top B with the deduplicated union emitted whole. A fused rank is not calibrated against any of these nulls and carries no tail-probability interpretation; it reports a selection, and each alert still carries the p-value of the detector that raised it.

Conformal calibration [6] is applied optionally and orthogonally to all of the above. It replaces each detector's model tail with that value's rank in the same detector's burn-in distribution, which is super-uniform whether or not the model is correct. It cannot change any single detector's own ordering, a rank transform being monotone, so per-detector figures are identical with and without it; what it changes is which detector supplies the minimum, and what a combination is summing. It also floors every p-value at $1/(n+1)$, so events more extreme than the whole burn-in sample tie there, and the model log p-value is retained as the tie-break.

4. The decision problem

R6 states that alert volume follows from a stated exchange rate rather than from a quota, and the reason is a pathology of quotas: a system tuned to emit 100 alerts a day emits 100 on a quiet day too, all of them benign. If nothing happened, the right number of alerts is zero.

We therefore score an alerting configuration by

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U = v·TP − c·FP,
```

where v is what catching one incident is worth and c what one wasted investigation costs. Only the ratio v/c is identifiable from counts, so an operator supplies one number. A queue is worth reading exactly when TP/FP > c/v, so a target true-to-false ratio *is* the exchange rate; but utility also says how many rows to take, which a ratio cannot.

Maximising a ratio does not work here, and the reason is worth stating because the failure is structural rather than a matter of tuning. TP/FP is precision under the monotone transform P/(1−P), so maximising it maximises precision, whose maximiser is the smallest queue containing a true positive. Forbidding TP = 0 moves that corner from "alert on nothing" to "alert on one thing" rather than removing it. Any objective that is a function of precision alone is scale-free, and a scale-free quantity cannot say how many rows to show; the objective must contain a term that grows with true positives found. The same argument disposes of accuracy, balanced accuracy, F1 and Youden's J as objectives at this base rate: each is a fixed and unstated exchange rate wearing a metric's clothes. Accuracy is the sharpest case — over the scored window, alerting on nothing scores 99.9987%.

A budget is therefore a **ceiling rather than a quota**: within it, the queue is truncated where marginal alerts stop paying. Section 5.5 measures what that truncation costs and buys, at an exchange rate of v/c = 10 stated in advance.

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5. Results

All comparisons are at matched alerts per analyst-day. Counts of alerts are over the whole seven-day window, so a budget of B corresponds to 7B alerts.

5.1 What distinguishes the campaign

Before any detector, it is worth asking what property of a labelled event a method could exploit. We therefore census five structural properties of an event relative to the history it was scored against. The properties are defined by facts about the event and the history — has this entity taken this value before, had this pair occurred before, is this hour improbable for this entity — and never by which detector produced the smallest p-value: a partition defined by our own detectors would flatter the framework by construction.

On `r11`, over 4,190,603 scored events of which 549 are labelled:

PROPERTY	LABELLED	SHARE OF LABELLED	BACKGROUND	SHARE OF BACKGROUND	LIFT
novel pairing for this entity	362	65.9%	19,841	0.473%	139
novel value for this entity	215	39.2%	8,913	0.213%	184
volume burst for this entity	199	36.2%	1,143,820	27.3%	1.3
outside this entity's hours	181	33.0%	1,576,171	37.6%	0.9
population-rare value	0	0.0%	4,835	0.115%	0.0

The properties are not mutually exclusive and 92 labelled events exhibit none of them.

Three readings, and they set up everything that follows.

The two per-entity novelty properties are enriched by factors of 139 and 184. That is where the discriminating information on this corpus is, and it is the empirical content of the claim section 1.2 makes on general grounds.

Population rarity is not merely weaker; it is **empty**. Not one of the 549 labelled events holds a value rare in the population, while 4,835 background events do. A test of population rarity on this campaign is therefore not a worse test of the same thing — it is a test of something the campaign does not exhibit. This is the strongest form the paper's central claim takes, and it is a statement about this campaign rather than about population-scope methods in general; section 5.3 shows the same property behaving differently on planted attacks.

The remaining two properties are **not enriched at all**: 1.3 and 0.9. Volume bursts and out-of-hours activity are as common among background events as among labelled ones, so a test of either cannot rank labelled events highly however well it is specified. That bounds what work on those two detectors could recover on this campaign.

It is not the whole explanation for their results, and section 6.2 gives the other half for each of them. The timing detector's p-value was floored by its own construction at a value at or above its own alert cut, so it could not alert at the two tighter budgets whatever it observed; section 6.2 reports the statistic that removes it, under which it reaches 1, 2 and 7 rather than 0, 0 and 6. The volume detector's tail is misspecified: on `r11`, 27,464 background events fall below 10^{-12} where a calibrated null over this many events predicts 4.2×10^{-6}, and no labelled event goes below 1.96×10^{-7}, so its queue is filled with background at every budget before a labelled event can compete. Section 6.2 shows a per-entity abstention accounts for none of it. The limits are real and independent of the enrichment result — a property that does not discriminate, and, separately, statistics that could not express discrimination if they had any.

5.2 Detection on the real campaign

`lanl-r11-b1000-weighted-d7-14-005` and `analysis-r11-b1000-conf-003`, 549 labelled events among 4,190,603 scored. Every proportion carries a 95% Wilson interval [7], and section 6.2 states why those intervals are optimistic.

ARM	10/DAY	100/DAY	1000/DAY	PRECISION AT 10/DAY	RECALL AT 1000/DAY
novelty	11	60	201	15.7% (9.0–26.0)	36.6% (32.7–40.7)
pairing	4	59	127	5.7% (2.2–13.8)	23.1% (19.8–26.8)
novelty rate	0	22	185	—	34.9% (31.0–39.1)
timing	1	2	7	1.4% (0.3–7.7)	1.3% (0.6–2.7)
volume	0	0	0	—	0.0% (0.0–0.7)
marginal	0	0	0	—	0.0% (0.0–0.7)
corrected minimum	4	47	162	5.7% (2.2–13.8)	29.5% (25.8–33.5)
composite	0	6	113	—	20.6% (17.4–24.2)

Four readings.

The per-entity novelty tests carry the campaign and the population test does not. At the tightest budget the novelty detector returns 11 true positives in 70 alerts: a precision of 15.7% (9.0–26.0) against a base rate of 1.31×10^{-4}, a lift of about 1,200. Precision falls to 2.9% at 1000 alerts a day while recall rises to 36.6%, which is the ordinary consequence of taking a longer prefix of one ranking rather than evidence about the ranking's quality.

Two arms are not distinguishable and should not be ranked. At 100 alerts a day the novelty detector catches 60 and the pairing detector 59, with recall intervals of 10.9% (8.6–13.8) and 10.7% (8.4–13.6). Those are the same measurement. The two questions are also nearly the same question — a novel pairing is a novel value of a composite field — and section 5.4 shows their detections overlap heavily.

The volume and marginal arms detect nothing at any budget, and section 5.1 says why. Their zeros carry an upper bound rather than a point: 0 of 549 is a recall of 0.0% (0.0–0.7).

Neither combination reaches its own best component at any budget. The corrected minimum reaches 4, 47 and 162 against the novelty detector's 11, 60 and 201; the composite reaches 0, 6 and 113. The composite's zero at 10 and 100 alerts a day is not a calibration artefact that calibration removes — those figures are with conformal calibration applied, and without it the composite reaches 0 at 100 a day rather than 6. Section 5.4 diagnoses both failures.

The same measurement on a corpus differing in background population (`holdout-r7-fixed` , 262 labelled among 1,427,225 scored) gives the novelty detector 2, 8, 15 and 21 at 10, 25, 50 and 100 alerts a day, and the composite 0 at every budget. Because labelled accounts are exempt from sampling, that corpus shares its labelled events with this one; it is a check on sensitivity to background, not an independent replication.

5.3 Detection by attack mechanism

The planted corpus separates "this test cannot express this mechanism" from "this budget cannot afford it". `lanl-inj-b1000-weighted-d7-14-005` : 856 planted events across six mechanisms plus the 549 real labelled events. Attribution is by victim account, which the design makes unambiguous. Planted and real ground truth are reported side by side and never summed.

At 10 alerts a day.

METHOD	SPRAY	LATERAL	OFF-HRS	PRIV-ESC	LOW+SLOW	TAKEOVER	REAL	ALERTS
per-entity detector								
`novelty`	0	0	0	0	0	0	11	70
`noveltyrate`	0	0	0	0	0	0	0	70
`pairing`	0	0	0	0	0	0	4	70
`timing`	0	0	0	0	0	0	2	70
`volume`	0	0	0	0	0	0	0	70
population detector								
`marginal`	0	0	0	0	0	0	0	70
combination								
composite (Fisher + Brown)	0	0	0	0	0	0	0	70
corrected minimum (Šidák)	0	0	0	0	0	0	4	70
union, per-entity arms (equal cost)	0	0	0	0	0	0	2	70
union, per-entity arms (equal depth)	0	0	0	0	0	0	13	294 **(×4.2)**
union, all arms (equal cost)	0	0	0	0	0	0	2	70
union, all arms (equal depth)	0	0	0	0	0	0	13	362 **(×5.2)**

At 100 alerts a day.

METHOD	SPRAY	LATERAL	OFF-HRS	PRIV-ESC	LOW+SLOW	TAKEOVER	REAL	ALERTS
per-entity detector								
`novelty`	0	0	0	0	0	0	60	700
`noveltyrate`	0	0	0	0	0	0	21	700
`pairing`	0	0	0	0	0	0	59	700
`timing`	0	0	1	0	0	5	3	700
`volume`	0	0	0	0	0	0	0	700
population detector								
`marginal`	0	0	0	0	0	76	0	700
combination								
composite (Fisher + Brown)	0	0	0	0	0	26	0	700
corrected minimum (Šidák)	0	0	0	0	0	0	47	700
union, per-entity arms (equal cost)	0	0	0	0	0	2	26	700
union, per-entity arms (equal depth)	0	0	1	0	0	5	77	2731 **(×3.9)**
union, all arms (equal cost)	0	0	0	0	0	2	22	700
union, all arms (equal depth)	0	0	1	0	0	76	77	3400 **(×4.9)**

At 1000 alerts a day.

METHOD	SPRAY	LATERAL	OFF-HRS	PRIV-ESC	LOW+SLOW	TAKEOVER	REAL	ALERTS
per-entity detector								
`novelty`	80	15	0	0	0	30	178	7000
`noveltyrate`	117	26	0	4	0	64	173	7000
`pairing`	0	0	0	0	0	12	130	7000
`timing`	0	0	3	0	0	11	7	7000
`volume`	0	0	0	0	0	0	0	7000
population detector								
`marginal`	0	0	0	0	0	**120/120**	0	7000
combination								
composite (Fisher + Brown)	0	4	0	0	0	116	107	7000
corrected minimum (Šidák)	40	10	0	0	0	28	156	7000
union, per-entity arms (equal cost)	4	0	2	0	0	22	105	7000
union, per-entity arms (equal depth)	117	26	3	4	0	71	265	25818 **(×3.7)**
union, all arms (equal cost)	0	0	2	0	0	**120/120**	99	7000
union, all arms (equal depth)	117	26	3	4	0	**120/120**	265	31505 **(×4.5)**

Planted: spray 320, lateral 40, off-hrs 64, priv-esc 24, low+slow 288, takeover 120, real 549.

Read a per-mechanism count as a coarse rate. Each mechanism has eight victims and a deterministic choice of planted values, so the events of one mechanism are not independent draws and the effective sample size is nearer eight than the event count. "120 of 120" is eight victims of eight.

The budget binds harder than the method. At 10 alerts a day no arm reaches any planted mechanism at all; the only detections anywhere are 11 real-campaign events to the novelty detector and 4 to the pairing detector. At 100 a day exactly one mechanism is reached. At 1000 a day five of six are. For four of the six mechanisms, therefore, the sentence "this framework does not detect it" is false at some budget and true at another, and only the budget changed.

No single method is best at more than one thing. At the widest budget the marginal takes every planted takeover and nothing else; the novelty-rate detector takes credential spray, lateral movement and privilege escalation; the novelty detector takes the real campaign. A reader looking for one row to deploy will not find one.

Low-and-slow is reached by no arm at any budget: 0 of 288, or 0 of 8 victims. It is the mechanism the volume predictive's over-dispersion is built to tolerate, so this is a confirmed prediction rather than an unexplained gap, and section 5.1 shows the volume-burst property is not enriched in any case.

The population marginal detects every planted takeover, and the reason is not the one the design intended to exclude. The corpus plants the most population-common value the victim has never used, precisely so that population rarity is held out; so the marginal's 120 of 120 requires an explanation, and the measurement supplies it. Median p-values for the marginal are 0.59 on credential spray and lateral movement, which substitute the destination computer, and 0.17 on privilege escalation, which substitutes the authentication type; on takeover, which substitutes destination, authentication type and logon type at once, the median is 3.8×10^{-5}.

The pattern follows the cardinality of the substituted field. The destination computer takes 3,535 distinct values in this corpus, so a common value the victim has not used is genuinely common and the marginal is unmoved. Authentication type takes 14 and logon type 10, with steep distributions — Kerberos, NTLM and Negotiate account for 99.6% of resolved authentication types — so an account that already uses the common values can only be given one from the tail. Holding population rarity out succeeds where the vocabulary is large and cannot where it is small, and the takeover row is where that shows. It is a limitation of the planted corpus, stated here because the alternative is to read the row as evidence for a mechanism the design was built to exclude.

5.4 Combination and budget allocation

Section 5.2 established that no rule tested reaches its best component. The two rules fail for different reasons and the union isolates them.

Fisher's sum averages an informative test with uninformative ones. Labelled events sit at the 0.07th percentile of the novelty detector's own distribution and between the 18th and 36th of every other detector's. Summing $-2 \ln P$ over six tests dilutes one signal with five non-signals. Fisher's method is powerful against diffuse alternatives and the minimum against sparse ones; this alternative is sparse.

The corrected minimum compares p-values across tests that share no scale. Under it the novelty detector supplies 5,979 of the 7,000 retained alerts, 85%. Four other detectors divide the remaining 15% — pairing 396, novelty rate 289, marginal 217, timing 119 — and two, volume and co-occurrence, never supply the minimum at all. A detector's p-value is a statement under its own null, so the arm whose null produces numerically smaller values takes the queue whatever the others found.

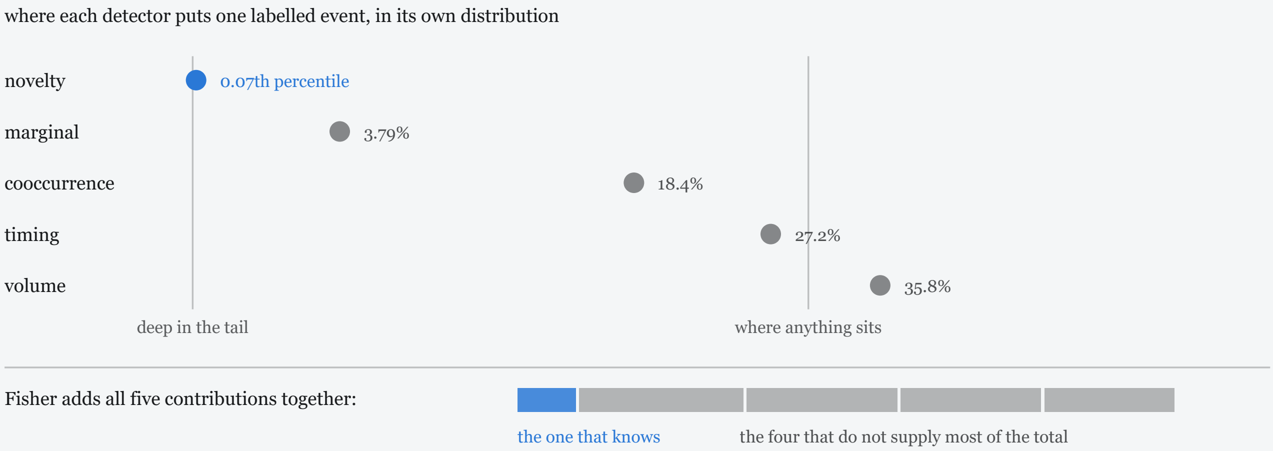
The union removes that scale, and the result separates two hypotheses that had been running together.

RULE	10/DAY	100/DAY	1000/DAY	ALERTS AT 1000/DAY
best single arm	**11**	**60**	**201**	7,000
union, per-entity arms, equal cost	2	27	106	7,000
union, all arms, equal cost	2	21	96	7,000
union, per-entity arms, equal depth	11	74	278	25,930 (×3.7)
union, all arms, equal depth	11	74	278	32,134 (×4.5)

At equal cost the union is strictly dominated at every budget — fewer labelled events and more false alarms than using the best arm alone. The intervals do not overlap: at 1000 a day recall is 17.5% (14.5–20.9) against 36.6% (32.7–40.7). No exchange rate rescues a dominated option.

At equal depth it finds what no single arm finds: 278 against 201 at 1000 a day, recall 50.6% (46.5–54.8) against 36.6% (32.7–40.7), which is a real and separated difference. It spends 3.7 times the alerts to get it.

So the scale mismatch was not the binding defect. Removing it made the marginal's signal reachable and the result at equal cost got *worse*: the novelty detector holding 84% of the minimum's queue was closer to correct than crowding-out suggested, because it is the better arm. What the union changes is the accounting, not the ranking. The diagnosis it leaves is that a fixed budget divided among six arms gives each about a sixth of the depth it had alone, and section 5.5 tests whether dividing it by quality instead recovers anything.



At 100 alerts a day, of 262 labelled events: novelty alone 21 · most-extreme-of-five 20 · **the five-detector sum 0**

More evidence is not better when most of it is not evidence. Fisher's method asks whether the evidence is jointly unusual, which is the right question only when every detector is informative. Here one is: it places the median labelled event at the 0.07th percentile of its own distribution, more extreme than 99.93% of all traffic. The other four place it where any event would sit. The sum is dominated by the four, so a composite built from a component that finds 21 labelled events finds none — and no cleverer combiner recovers it, because the remedy is not a better average but not averaging in detectors that know nothing.

5.5 Allocation by demonstrated quality, and why it cannot help

Section 5.4 leaves one diagnosis standing: every rule tried divides a fixed budget by quota rather than by quality, so a detector that finds nothing draws the same share as the best one. This section tests that diagnosis and rejects it.

The rule measured is a per-alert likelihood ratio. Two quantities are fitted per detector on the burn-in window and frozen at the boundary: a null over that detector's own log p-value, and the single parameter a of a Beta(a, 1) density over the null quantiles its labelled burn-in events received. An alert at quantile q from a detector of weight a scores

```
s = ln a + (a − 1) ln q,
```

the log density of q under that Beta and, because a null quantile is uniform under the null by construction, the log-likelihood ratio of the alert being labelled against its being background. Alerts from detectors sharing no p-value scale are comparable on s, so the budget goes to the highest scores and no share parameter is chosen. Two details of the fit matter. A weight is retained only where twice its log-likelihood ratio against $a = 1$ exceeds 2.706, the 5% one-sided point for a parameter tested at the boundary of its range [10]: without it, fifty uniform draws fit $a \approx 1 \pm 0.14$, and at $a = 0.93$ the ratio scores an alert at $\ln q = -4000$ some 248 log units above zero, so noise alone would buy a large share of the queue. And labelled events a detector evaluated without surfacing enter as right-censored observations, without which a detector that surfaced two of forty-nine at its two most extreme ranks is fitted as the sharpest in the set.

The fit separates the detectors in the expected direction. On `r11` the three per-entity arms are informative — a = 0.376, 0.414 and 0.491 for the novelty, novelty-rate and pairing detectors — while the timing, volume and marginal detectors surface none of the 49 labelled burn-in events, are fitted uninformative, and therefore cost nothing. That is the intended behaviour, and it is not enough.

RULE	`R11` 100/DAY	`R11` 1000/DAY	`INJ` 100/DAY	`INJ` 1000/DAY
best single arm	**60**	**201**	**76**	**384**
weighted, burn-in weights	56	164	56	231
weighted, oracle weights	61	174	61	309
best two-arm split, oracle	60	201	77	395

`lanl-r11-b1000-weighted-d7-14-005` and `lanl-inj-b1000-weighted-d7-14-005`. The two oracle rows read the labels they are evaluated against; neither is a deployable configuration, and both are here to bound what a fitted rule could reach. Three of the four rows are computed by these runs from the same rankings; the oracle-weights row is not recorded by the replay and was computed under the timing statistic section 6.2 replaces.

The burn-in-weighted rule loses at every budget on both corpora. That alone would leave the diagnosis open, because 49 labelled events is a thin sample for ranking six detectors, and a rule may be sound while its estimator is starved. Refitting the same rule's weights on the evaluation labels settles it. Those weights separate the arms properly — 0.19, 0.22 and 0.23 for the three per-entity arms against 0.71 for timing and 1.0 for volume — and the rule still loses, by 27 detections on the real campaign at the widest budget and by 75 on the planted corpus. The construction is what fails, not the fit.

An exhaustive search over two-arm splits, again choosing the split with the labels in hand, shows why. On the real campaign the optimum is the corner: the whole budget to the best single arm, at both budgets. Diverting 5% of it costs 13 detections at 1000 alerts a day, so the derivative is negative *at* the corner and no allocation over any number of arms can improve on it.

The mechanism is that the arms are substitutes rather than complements. At 1000 alerts a day on the real campaign, 74.6% of the novelty-rate detector's detections are also found by the novelty detector, and 78.7% of the pairing detector's are. Splitting a budget halves each arm's depth, and what survives the halving largely coincides, so the union of two half-depth prefixes is smaller than one full-depth prefix. The 77 additional detections the equal-depth union reaches (section 5.4) are real, and they are reachable only by spending more alerts, not by spending the same alerts differently.

Where the arms are genuine complements the same search finds the headroom, which is the control this argument needs. On the planted corpus at 100 alerts a day the population marginal detector's 76 detections overlap the per-entity arms' by **zero** — the two scopes answer different questions and, there, catch disjoint events — and the best split gives the marginal 75 of the 100 daily alerts and the novelty detector 25, for 77 against 76. At 1000 a day it gives the marginal 150 and the novelty-rate detector 850, for 395 against 384. Both optima sit on a plateau — several splits reach the same count — which is itself a sign of how little is at stake. Both gains are inside the sampling error of the counts, and both are on the corpus of planted attacks rather than the real campaign.

So the overlap between two arms' detections is what decides whether dividing a budget can pay, and on this corpus no division of a fixed budget beats using the best detector alone. The reason is not that the wrong rule was tried.

One property of the rule survives its negative result, and is why the implementation retains it. Every quantity s reads is a property of the single alert or of state frozen before the scoring window began, so the same arithmetic thresholds a stream in service against the operating point section 4 derives. The null is deliberately not a rank in the burn-in sample: a rank cannot fall below $1/(n+1)$, the alerts worth having are past that floor, and a rank therefore ties the head of the queue and orders it by arrival. Past a high threshold the null is extended by an exponential fit to the burn-in excesses, monotone everywhere and linear in the log p-value, so nothing ties and nothing underflows at $\ln P = -4000$.

5.6 Truncating where the objective peaks

At the exchange rate v/c = 10 stated in section 4, and on the corrected-minimum arm:

BUDGET	PERMITTED	TP	OPTIMAL	TP	PRECISION	SUPPRESSED	TP FORGONE
10/day	70	4	**30**	4	5.7% → **13.3%**	40 (57%)	**0**
100/day	700	47	**221**	47	6.7% → **21.3%**	479 (68%)	**0**
1000/day	7,000	162	**315**	74	2.3% → **23.5%**	6,685 (95%)	88

At 10 and 100 alerts a day the truncation is free: 57% and 68% of the queue goes for no loss of detection. At 1000 it becomes a decision — the objective discards 6,685 alerts and 88 true positives with them, because at this rate those 88 do not pay for 6,685 wasted investigations — and precision rises tenfold. For the composite the optimum is to emit nothing at any budget, including the budget at which it finds 113 true positives, which do not pay for 6,887 false ones at v/c = 10. An objective declining to deploy a detector is a usable answer.

This is an oracle bound: the optimum is located using the labels, so it measures the headroom a cutoff has rather than being a deployable rule. Section 6 says what a deployable version needs.

Reporting at a nominal false discovery rate instead is not available on this corpus. Applying Benjamini–Hochberg [8] per day to the composite at q = 0.05 yields 5,352 discoveries of which 113 are labelled, a realised FDR of 0.979 (0.975–0.982), with five of seven days saturated; Benjamini–Yekutieli [9] under dependence gives 0.978 at q = 0.001. The nominal level has no purchase because the p-values it is applied to are not calibrated, and that is a property of the composite rather than of the procedures.

5.7 Population-scope baselines

Six population-scope reference implementations spanning four inductive biases were run: isolation forest [11] and extended isolation forest [12] from scikit-learn [16], half-space trees [13], local outlier factor [14], one-class SVM [15] and PCA reconstruction error. The one-class SVM is a Nyström kernel map with a stochastic-gradient one-class objective rather than the exact kernel machine, which is $O(n^2)$ in the fit set and cannot be run at this row count; it is a deviation from the textbook formulation and is recorded as one. A per-entity reference, an uncalibrated EWMA z-score, was also run.

METHOD	SPRAY	LATERAL	OFF-HRS	PRIV-ESC	LOW+SLOW	TAKEOVER	REAL	ALERTS
baseline (per-entity)								
`entity_ewma`	0	0	0	0	0	1	0	--
baseline (population)								
`eif`	0	0	0	0	0	0	0	--
`hst`	0	0	0	0	0	0	0	--
`iforest`	0	0	0	0	0	0	0	--
`lof`	0	0	0	0	12	0	0	--
`ocsvm`	0	0	0	0	0	33	0	--
`pca`	0	0	0	0	0	0	0	--

On the matched comparison (`baselines-holdout-r7-d7-9` , the same 262 labelled events section 5.2 reproduces on) all six population models detect **0 at every budget**, which section 5.1's census explains: not one labelled event of the real campaign holds a population-rare value.

On the planted corpus two of them are not zero, and the exception is the interesting one. One-class SVM detects 33 of 120 planted takeovers and local outlier factor 12 of 288 low-and-slow events — the one mechanism no arm of this framework reaches at any budget. A sustained small departure is what the volume predictive's over-dispersion is built to tolerate and what a density estimate has no such tolerance for. That is a capability this framework does not have, and it is why sections 1.3 and 6 describe the two scopes as complementary rather than ordered.

Both baseline figures come from an easier problem, in one way that dominates: the models read a 1-in-100 event sample of 45,071 rows rather than 4.49 million events. That raises the labelled share from 0.031% to 3.1%, and it raises the share of the corpus a fixed budget covers by the same factor of about 100. Given a hundredfold easier problem they reached 33 and 12.

The per-entity reference cannot be read at all from this run, and we decline to read it. Sampling uniformly over events decimates every entity's history at the same rate, so an EWMA over an entity's own past is estimated from a hundredth of that past; the export records `entity_history_intact: false`. Its single detection is neither evidence for nor against the per-entity framing, and a matched entity-sampled export is needed before it means anything.

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6. Discussion

6.1 What the evidence supports, and what it does not

Conditioning each null on the entity that produced the event is supported on this campaign, and the support is specific: the properties per-entity tests express are enriched 139- and 184-fold among labelled events, and the property a population-rarity test expresses contains none of them. The direction of that comparison is what the evidence establishes. It is narrower than "per-entity conditioning beats population conditioning", because on planted attacks the population test detects a mechanism no entity-scope test here reaches, and a population density estimate reaches a second one.

Combining evidence across the tests is not supported. Three rules were measured and none exceeds its best component at equal cost at any budget, with non-overlapping intervals at the widest budget. The union at equal depth does exceed it, and costs 3.7 times the alerts to do so; whether that trade is worth taking is what section 4's objective decides and depends on an exchange rate an operator must supply.

6.2 Threats to validity

The novelty p-value is confounded with history length. For a first-ever value the estimate reduces to the reserved mass $a/(n + a(K+1))$, approximately $1/n$. Across 32 planted victims spanning 263 to 20,666 events of history, the product of the p-value and the history length has median 1.15 and lies in [0.50, 7.22]: among novel events the ranking is close to sorting accounts by event count. Clearing the detector's realised cut on this corpus requires of the order of 10^5 events of history, and the busiest planted victim had 20,666, so **no attack on an ordinary account could win an alert slot regardless of what it did**. This is a covariate problem — history length is informative about the p-value's scale while being no part of the question — and it is the single largest methodological limitation here. It is also not straightforwardly fixable: the one principled correction tried, reserving unseen mass by Good–Turing rather than by a fixed concentration, *lost* detections, because large histories carry many singletons and the correction therefore judges a first-ever value unremarkable for exactly the accounts the working estimator ranks highest. The working configuration works for a reason the model does not state, and neutralising the covariate risks destroying the accidental correlation the result rests on.

A sustained intrusion enlarges its own reference set. State is committed after scoring, so an event is judged against a history that does not yet include it — but the next event of the same campaign is judged against a history that does. Over a campaign persisting for days on one account, the reference distribution drifts toward the attacker's behaviour. This is an untested alternative explanation for two of the paper's null results, the low-and-slow zero in particular, and it cannot be separated from the stated mechanism without a run that freezes per-entity state across the campaign window.

The timing detector's tail mass was floored by its own grid, at its own alert cut. It is read from a 512-point lookup over the entity's circular density and reports nothing below one half-cell, $1/(2 \times 512) = 9.77 \times 10^{-4}$, while the realised cut on the planted corpus is 3.98×10^{-3} at 1000 alerts a day and 1.00×10^{-3} at 100 and at 10. At the tighter two the most extreme possible answer was therefore at or above the cut: the detector could not alert whatever it observed, and its zeros were a property of the statistic rather than of the mechanism.

It was meanwhile responding to the mechanism it was built for, so this was a ceiling and not blindness, and raising the grid would not have lifted it: only 1 of 64 planted off-hours events and 5 of 549 real labelled events sat at the floor. The rest were held up by the statistic itself, because a tail mass over density levels saturates — for an account with an eight-hour working window, an event on the opposite side of the clock still sits in a low-density region covering much of the circle.

The statistic is therefore now the event's ln U standardised by the mean and spread of the ln U this entity's own events have received, which has no floor at all. It costs two numbers of state per entity, and the detector abstains where that history is too short to estimate the null or, for a perfectly regular account, carries no spread to standardise against: 4,847 events of 4,190,603. Detections at 10, 100 and 1000 alerts a day rise from 0, 0 and 6 to 1, 2 and 7 on the real campaign, and from 0, 1 and 12 to 2, 9 and 21 on the planted corpus. The off-hours response is not traded away for that: the median p-value on planted off-hours attacks falls from 3.20×10^{-2} to 8.23×10^{-3}, widening the separation from the nearest other planted mechanism from 5.7 to 18.6. Every other arm is unchanged on both corpora.

The volume detector now abstains where an entity has no completed period, as R3 requires: with none the rate posterior of equation (10) is the prior, and reporting that as P = 1 is an opinion. On `lanl-inj-b1000-weighted-d7-14-005` it abstains on 4,399 of 4,494,396 events.

It does not repair the queue. 13,618 events on the planted corpus and 27,464 on `r11` still fall below 10^{-12}, where a calibrated null predicts about 4×10^{-6}; no labelled event falls below 1.96×10^{-7}; and the realised cut stays on that floor on six of seven scored days at every budget, so volume detects nothing at any budget on either corpus. One, two, three and five completed periods were each measured from a single ungated pass (`results/volume-abstention-gate.json`) and none clears the floor: a first period scores P = 1 exactly, so a gate at one removes none of the pile, and at five — 4.9% of events abstained, 132 labelled events withheld — only 10.6% of it goes. The pile is made of entities with established history, so the cause is equation (11)'s predictive being too narrow for their habitual variation, not the cold start. Its response is inverted as well — median p 0.72 on planted low-and-slow attacks against 0.29 on the other mechanisms — a second defect the abstention does not touch.

Neither this detector nor the timing detector is therefore evidence about per-entity conditioning in either direction, and section 5.3's zeros for them should be read as unmeasured rather than as measured absences. Correcting either changes every figure they appear in, so both are reported here rather than patched.

Effective sample size is far below nominal, so every interval here is optimistic. The 549 labelled events fall on 104 accounts, 93.6% of label rows share one source computer, and two of seven days carry most of them; the unit of independence is nearer the account or the campaign-day than the event. On the planted corpus each mechanism has eight victims and a deterministic choice of values. The Wilson intervals treat alerts as independent draws and therefore understate uncertainty; they are reported because a count with no interval understates it further.

Labels are a partial record, so the figures are bounded in known directions. Malicious activity the red team did not record counts against precision, so every precision figure is a lower bound; and the recall denominator is what was written down. Separately, failed authentications appear in this corpus only for accounts that succeeded somewhere in it, so the failure population is conditioned on eventual success — a property of the archive, not of a live stream, and one the detectors score.

Several quantities are chosen on the outcome. "The best single arm" is identified per corpus after seeing labels; the exchange rates at which the equal-depth union becomes preferable are computed from the same labels; the truncation optimum in section 5.6 is located using them and is stated as an oracle bound. Across the study, seven detectors, four combination rules, two costings, three budgets, two label sets and two corpora were evaluated with maxima reported, and no multiplicity adjustment is applied to that outer loop. The burn-in split is the one thing fixed in advance, and it is recorded by the commit that fixed it. A held-out corpus for the selected configurations is the correct remedy and has not been run.

Nothing here is measured at full population. Every figure is from an entity subset, where a fixed alerts-per-day budget is a far easier target than on 42.2 million events. One full-population run exists; it scored 42,218,530 events with four of the seven detectors and reports its composite detecting nothing at any budget, so the per-entity arms have never been measured at full scale. A detection rate measured on a subset is not comparable to one measured on the whole.

The negative variance estimate is a finding about the composite, not a nuisance. With six detectors the burn-in covariance implies $\mathrm{Var}[X^2] = -27.5$. Brown's correction absorbs dependence between test statistics; a variance no joint distribution can produce says the estimate is measuring the marginals' misspecification rather than the detectors' dependence. Two of the nulls are known not to be calibrated, which is consistent with that reading. It also means the composite figures throughout are plain Fisher, and that a corrected composite on this corpus awaits calibrated marginals rather than a better covariance estimator.

Aggregating to the entity-day raises apparent recall, and the confound accounts for most of it. An analyst triages accounts rather than log lines, and the same evidence aggregated to (entity, day) reaches 65 of 108 labelled entity-days at 100 alerts a day against 47 of 549 labelled events at the same budget. But Fisher's statistic at entity scope grows with the event count, so the ranking sorts by activity as much as by anomaly; replacing it with the count-normalised standardised form takes 65 down to 14. Most of the apparent gain is the confound. Reporting the aggregate without that correction would be the most flattering presentation available in this paper, which is why it is here rather than in section 5.

6.3 What a deployable version needs

One gap separates what is measured here from something that can be run: a fixed budget is a batch construction. Selecting the top B of a day requires the whole day, and an operator at 14:00 does not know what arrives by 23:59; the same objection applies to a per-day Benjamini–Hochberg step-up, whose threshold depends on the number of tests. Section 4's objective does not have this defect, comparing one alert against a stated exchange rate and needing no day context, and section 5.5's score is built to the same constraint. Reporting at matched budgets is how this paper charges methods a comparable cost, not a claim about how a threshold should be set in service. What remains missing is a calibrated per-alert probability: section 5.6's truncation uses labels to place alerts on one scale and is therefore a bound rather than a rule.

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Data and code availability

Every number in this paper is read from a machine-generated result file committed alongside the source, each carrying its run identifier, corpus checksums, row counts, seeds and parameters. No figure is a plot of data: each is a diagram of a mechanism, so no figure can disagree with a result. The corpus is third-party and is not redistributed; its provenance and licence are recorded with the code.

Generated by `cmd/thesis` from `docs/PAPER.md`, which stays canonical for the prose: this page is never edited directly, so the two cannot disagree. Every quantitative claim traces to a recorded run in `results/`. Diagrams are hand-authored inline SVG with no external resources, and their two carrying colours are validated for contrast and colour-vision separation against both the light and dark surfaces.